

[5353]-196

T.E. (Information Technology)**DESIGN AND ANALYSIS OF ALGORITHMS****(2012 Pattern)***Time : 2½ Hours]**[Max. Marks : 70**Instructions to the candidates:*

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Neat diagram must be drawn whenever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

Q1) a) Explain direct proof & proof by counter example techniques. **[4]****b)** Define following asymptotic notations : **[6]**

- i) Theta
- ii) Little oh
- iii) Little Omega

OR**Q2) a)** Setup a recurrence relation to compute $n!$ and solve it. **[5]****b)** Consider the following letters with their probability. **[5]**

Character	a	b	c	d	e
Probability	0.5	0.25	0.125	0.625	0.031

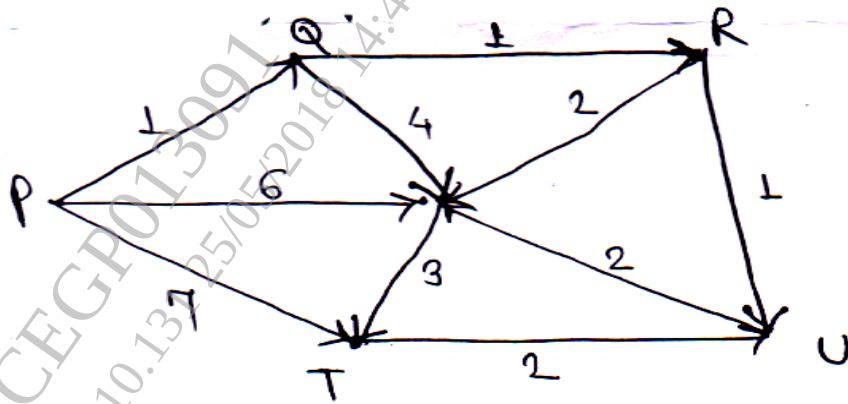
Find out Huffman coding for a, b, c, d, e.

Q3) a) Show the steps in multiplying the following two integers using efficiency integer multiplication method 2101×1130 . **[5]****b)** Explain the following terms with reference to Greedy Technique **[5]**

- i) Feasible solution and optimal solution
- ii) Subset paradigm and ordering paradigm

OR

- Q4) a)** Apply Dijkstra's algorithm for finding all shortest paths from a single source 'p' in a given graph [5]



- b) Explain the concept of divide and conquer technique. Write master theorem. [5]

- Q5) a)** Let $w = \{5, 10, 12, 13, 15, 18\}$, $m = 30$. Find all possible subsets of w that sum to m . Draw the portion of state space tree that is generated. [8]
- b) Write a recursive backtracking algorithm for m -coloring of the graph. [8]

OR

- Q6) a)** What is m -colorability optimization problem? Explain with an example. [8]
- b) Find all possible solutions for 5 queens' problem using backtracking method. [8]

- Q7)** What is travelling salesman problem? Find the solution of following travelling salesman problem using branch and bound method. [18]

∞	20	30	10	11
15	∞	16	4	2
3	5	∞	2	4
19	6	18	∞	3
16	4	7	16	∞

Cost Matrix

OR

- Q8)** a) What is LC Search? Explain in detail control abstraction for LC search. [8]
b) Solve the following instance of 0/1 knapsack problem by FIFO branch and bound approach. $n = 4$, $M = 15$ and $(P_1, P_2, P_3, P_4) = (10, 10, 12, 18)$; $(W_1, W_2, W_3, W_4) = (2, 4, 6, 9)$ [10]
- Q9)** a) What do you mean by P, NP, NP-Hard and NP-complete problems? Give an example of each category. [8]
b) Explain the need and significance of parallel algorithms. [8]

OR

- Q10)** a) Explain the following : [8]
i) Computational complexity
ii) Decision problems
iii) Deterministic and Non - deterministic classes
iv) Complexity classes
b) Explain the possible parallel computations using complete binary tree. [8]

